

CAEM: Final Research Plan

Confidence-Aware Episodic Memory with Self-Improvement

Revised April 2026 — Post EXP-19, Cloud GPU Confirmed

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Contents

1 Status Summary

What is complete:

- Chapters 1–2: submitted (Phase 1 pre-thesis report).
- Chapter 3 (Requirements, Impacts, Project Management): written and compiled.
- Chapter 4 (Methodology): written, all 5 algorithms in **algorithmic** style, all figures placed, notation table, theoretical analysis including Data Purity Theorem.
- Implementation: all 8 pipeline stages, 96 unit tests passing, mini-run (EXP-19) completed on RTX 3060 (n=500, 4 cycles).
- Mini-run results confirmed: FEVER +10.9%, StrategyQA +9.4%, TruthfulQA net +4.0%, HotpotQA EM≈0 (boundary condition).

What is pending:

- FEVER data leakage fix (Section ??) — **blocking**.
- Dataset suite revision (Section ??).
- Full-scale cloud experiment (Section ??).
- Ablation suite (Section ??).
- Chapters 5 and 6 (Section ??).

2 Critical Fix: FEVER Data Leakage

Data integrity issue — fix before any full-scale run.

Problem: `run_experiment.py` loads FEVER from the `train` split of `lucadiliello/fever` for both the self-improvement loop (SIL) query pool *and* evaluation. Even though `split_calibration_sets()` carves out calibration and purity subsets, the evaluation queries are drawn from the same `train` split. The model is therefore evaluated on queries from the same distribution it was trained on — a direct data leakage. Mini-run FEVER results (+10.9%) are *potentially inflated* and cannot be reported as scientifically valid until the fix is applied and results re-checked.

Root cause: `lucadiliello/fever` has three splits:

- `train` — 145,449 examples (labels available).
- `paper_dev` — 19,998 examples (labels available) — the standard published benchmark split used by every FEVER baseline (DPR, FiD, Atlas).
- `paper_test` — unlabelled.

`paper_dev` was never referenced in `run_experiment.py`.

Fix:

1. Evaluation for FEVER must use `paper_dev` exclusively. Never pass `paper_dev` examples to the SIL query pool.
2. `train` is allocated as follows (in order, no overlap):

- Indices 0–499: calibration set (for $\hat{u}$ logistic regression).
 - Indices 500–999: purity validation set (for verifier accuracy α).
 - Indices 1000+: SIL query pool (fed into the self-improvement loop).
3. Update `scripts/run_experiment.py` and `caem/benchmarks/fever.py` to load `paper_dev` for evaluation.
 4. Rerun mini-run smoke test with fix applied to confirm FEVER results are still positive (expected: similar magnitude, but now scientifically clean).

Secondary benefit: Using `paper_dev` as the eval set makes CAEM’s FEVER numbers directly comparable to every published baseline that also uses `paper_dev`, including the Liu et al. (ACL 2024) RA-ISF numbers cited in PUB-01.

3 Final Dataset Suite

3.1 Removed Datasets

HotpotQA — removed entirely. EXP-19 confirmed Flan-T5-Large achieves $EM \approx 0.006$ on HotpotQA across all four cycles. The Data Purity Theorem’s precondition $p > (1 - \alpha)$ is not satisfied when base accuracy is near zero. This result is reported as a boundary condition validation in Chapter 5, using mini-run data (no further HotpotQA compute needed). Do not include HotpotQA in the full-scale experiment.

3.2 Training Datasets (Self-Improvement Loop)

These datasets feed the SIL query pool across all cycles. Selection criteria: (a) sufficiently large training split for 10 cycles at 5,000 queries/cycle without repetition; (b) Flan-T5-Large has non-trivial base accuracy ($p \gg 0$), satisfying the purity theorem precondition; (c) verifiable answers compatible with NLI + SC + SE signals.

Each training dataset provides 5,000 queries per cycle. At 10 cycles, each dataset contributes 50,000 total queries — well within its training split without repetition.

Why not StrategyQA as training? StrategyQA has only 2,290 training samples. At 5,000 queries/cycle, the pool is exhausted before Cycle 1 completes. Repeated passes risk data memorisation rather than self-improvement. StrategyQA is re-assigned to the transfer evaluation suite (Section ??).

3.3 Evaluation Datasets (Transfer / Held-Out)

These datasets are *never* seen during training or the SIL loop. They are evaluated post-cycle to measure out-of-distribution generalisation.

Transfer narrative: If CAEM trained on FEVER + TriviaQA + Natural Questions improves on TruthfulQA, StrategyQA, and ARC-Challenge — benchmarks it never trained on — the system demonstrates generalised hallucination reduction, not just in-distribution optimisation. This is a stronger claim than improving on the training benchmarks alone.

Table 1: Self-improvement loop training datasets (final).

Dataset	Train size	Cycles covered	Rationale
FEVER	145,449	29×	Fact verification; NLI maps directly onto task; primary verifier signal. Fix leakage before use (Section ??).
TriviaQA	78,785	15×	Factoid; diverse domains; Flan-T5-Large has reasonable base capability. Unique in testing parametric knowledge breadth rather than reasoning chains.
Natural Questions	79,168	15×	Real user queries; open-domain; large; diverse. Complements FEVER (verification) and TriviaQA (trivia) with grounded natural-language questions.

Table 2: Transfer and held-out evaluation datasets (final).

Dataset	Eval size	Role and rationale
TruthfulQA	817	Adversarial misconceptions; tests whether hallucination reduction on factoid/verification tasks transfers to misconception resistance. Evaluation only — no training. Primary transfer claim.
StrategyQA	490 (test)	Implicit multi-step reasoning; tests transfer to reasoning tasks not in the training distribution.
ARC-Challenge	1,172	Science reasoning; hardest tier of ARC benchmark; tests transfer to domain-specific factual inference beyond Wikipedia.
HotpotQA (boundary)	500	Mini-run data only. Reports theorem boundary condition ($p \approx 0 \Rightarrow$ no improvement). No new compute.

Table 3: Final experimental configuration. All values are design decisions ([DES]) unless marked [LIT] (literature-fixed) or [CAL] (empirically calibrated).

Parameter	Value	Rationale
Base model	Flan-T5-Large (780M) [LIT]	Unchanged. Instruction-tuned; multi-task capable; data purity theorem boundary condition validated on this scale.
# self-improvement cycles	10 [DES]	Theory predicts practical equilibrium at 7–10 cycles. 10 cycles captures full convergence curves for all three training benchmarks.
Queries per benchmark per cycle	5,000 [DES]	Sufficient for statistically meaningful per-cycle EM; no dataset exhaustion across 10 cycles given training split sizes.
Max episodic memory	1,000,000 episodes [DES]	Removes capacity as a confounding variable. Pruning trigger at 95% (950,000) retained as safety ceiling. FAISS IVF-PQ at 1M scale: ~1.5 GB for compressed vectors.
FAISS index type (memory)	IVF-PQ, <code>nlist=4096</code> , <code>M=64</code> , <code>nbits=8</code> [DES]	Handles 1M-scale retrieval efficiently within A100 VRAM.
RAG passage corpus	Full DPR Wikipedia [LIT]	21,015,324 passages (100-word chunks, Karpukhin et al. 2020). Standard split used by all published RAG baselines (DPR, FiD, Atlas) — ensures comparability. Needs ≥ 128 GB system RAM for CPU FAISS index.
RAG FAISS index type	IVF-PQ, <code>nlist=65536</code> [DES]	Scales to 21M passages; ~40 GB compressed.
Embedding model	<code>all-mpnet-base-v2</code> [LIT]	384-dim output. Unchanged.
Verification NLI model	RoBERTa-Large-MNLI [LIT]	Unchanged.
L2 regularisation λ	0.01 [DES] ₅	Adapted from Kirkpatrick et al. (2017); validated in EWC ablation (Section ??).
Fine-tuning mix	90% verified / 10%	Unchanged.

4 System Configuration (Final)

4.1 Memory at 1M Scale — Practical Notes

FAISS IVF-PQ at 1M episodes: Compressed vectors at $M=64$ bytes/vector: $1,000,000 \times 64 = 64$ MB. Centroid table: $4096 \times 384 \times 4 = 6$ MB. Total in-memory footprint: ≈ 70 MB — trivially within A100 VRAM alongside both models (~ 4 GB fp32 total).

Retroactive re-verification cost at 1M: Re-verifying all stored episodes at the end of each cycle scales linearly. Apply the delta-threshold optimisation: only re-verify episodes where $\hat{u}_{\text{stored}} \in [0.45, 0.70]$ (the uncertainty band). Episodes with $\hat{u}_{\text{stored}} > 0.70$ are stable; episodes with $\hat{u}_{\text{stored}} < 0.45$ are pruning candidates regardless. This reduces the re-verification pass to $\approx 30\text{--}40\%$ of stored episodes per cycle.

Full Wikipedia DPR index: The standard DPR Wikipedia index (21M passages) is available as `wiki_dpr` on HuggingFace or via the original DPR repository. Build once; the index is static across all cycles. Requires ≥ 128 GB system RAM for CPU-resident FAISS search during inference (passage retrieval does not require GPU). Cloud instances with 192 GB RAM (e.g. `a2-highgpu-1g` on GCP or `p3.8xlarge` with CPU offload on AWS) are adequate.

5 Experimental Design

5.1 Data Splits (Non-Overlapping — Critical)

5.2 Self-Improvement Loop Execution

1. **Cold-start seeding:** Seed episodic memory with ~ 200 verified episodes per training benchmark (600 total) from the SIL query pool before Cycle 0. This prevents empty-memory degradation in Cycle 1.
2. **Per-cycle loop** (repeat for cycles 0–9):
 - a. Sample 5,000 queries per training benchmark from the SIL pool (without replacement across cycles).
 - b. Run each query through the full 8-stage pipeline (encode $\rightarrow$ retrieve $\rightarrow$ confidence $\rightarrow$ route $\rightarrow$ generate $\rightarrow$ verify $\rightarrow$ store).
 - c. Collect verified episodes into episodic memory.
 - d. At end of cycle: retroactive re-verification of uncertainty-band episodes ($\hat{u}_{\text{stored}} \in [0.45, 0.70]$).
 - e. Fine-tune on verified episodes (90/10 mix, L2 regularisation, $\lambda = 0.01$).
 - f. Check forgetting abort: if MMLU/HellaSwag/BoolQ/GSM8K retention < 0.93 , revert to previous cycle weights.
 - g. Evaluate on all benchmark eval splits (training + transfer). Save per-question outputs for ablation analysis.
3. **Calibration:** After Cycle 0 completes, fit $\hat{u}$ logistic regression weights on the 500-sample calibration sets. Update `calibration/calibrated_config.json`. Re-run Cycles 1–9 with calibrated weights.

Table 4: Data split allocation. Splits must be non-overlapping across all uses.

Purpose	Size	Source split	Notes
<i>FEVER</i>			
Calibration	500	train[0:500]	Fits $\hat{u}$ logistic regression weights.
Purity validation	500	train[500:1000]	Measures verifier accuracy α .
SIL query pool	train[1000+]	train[1000:]	Fed into self-improvement loop.
Evaluation	19,998	paper_dev	Never seen during training. Standard benchmark split.
<i>TriviaQA</i>			
Calibration	500	train[0:500]	
Purity validation	500	train[500:1000]	
SIL query pool	train[1000+]	train[1000:]	
Evaluation	validation split	validation	Official held-out validation set.
<i>Natural Questions</i>			
Calibration	500	train[0:500]	
Purity validation	500	train[500:1000]	
SIL query pool	train[1000+]	train[1000:]	
Evaluation	validation split	validation	Official held-out validation set.
<i>Transfer evaluation (no training split used)</i>			
TruthfulQA	817	validation	Entire dataset for eval.
StrategyQA	490	test	Official test split.
ARC-Challenge	1,172	test	Official test split.

5.3 Baselines

All baselines are evaluated once (no self-improvement loop) and reported in Table 5.1 (main results).

Table 5: Baseline methods for Chapter 5 comparison.

Baseline		What it lacks vs. CAEM
Zero-shot	Flan-T5-Large	No memory, no verification, no self-improvement.
Chain-of-Thought	(3-shot)	No memory, no verification, no self-improvement.
RAG	(full Wikipedia)	No memory accumulation, no confidence routing, no self-improvement.
Self-Consistency	(M=3)	No memory, no routing, no self-improvement. Verification only.
Vanilla Fine-Tuning		No memory, no confidence gating, no retroactive cleaning. Trains on all generated data regardless of quality.
Memory-Only	(no FT)	No fine-tuning. Tests whether memory accumulation alone (without model improvement) explains gains.
GPT-3.5 vanilla		Published numbers from Liu et al. ACL Findings 2024 (RA-ISF, arXiv:2403.06840). No compute required.
GPT-3.5 + RAG		Published numbers from same source.
Self-RAG 13B		Published numbers from same source. Key comparison: CAEM at 780M vs. Self-RAG at 13B.

6 Ablation Studies

6.1 Internal Ablations (run_ablation.py — already implemented)

Nine configurations grouped by the mechanism they probe. Run after the full-scale experiment using saved per-question outputs.

6.2 Extended Ablations (cloud GPU — new experiments)

These were previously compute-blocked on RTX 3060. All are now feasible on A100 80 GB. Merge PUB-04 and PUB-05 into a single 4-variant fine-tuning run to avoid running two separate training sweeps.

Table 6: Internal ablation configurations (Chapter 5, Table 5.5).

ID	Config	Committee question answered
<i>Memory quality group</i>		
A1	No retroactive re-verification	Is memory quality improvement bidirectional, or does it only flow forward?
A-NLI	Single-layer NLI only	Do SC and SE add value beyond NLI alone?
A-verify	No verification (store all)	Does the verification gate prevent memory contamination?
<i>Routing group</i>		
A2	All-Tier-3 (no routing)	Does the routing architecture provide a computational dividend?
A3	No OR-condition override	Does the safety override prevent routing failures?
<i>Training group</i>		
A4	Vanilla fine-tuning (no L2)	Does L2 regularisation prevent catastrophic forgetting?
A5	No fine-tuning (memory only)	Is fine-tuning necessary, or does memory accumulation alone explain gains?
<i>Confidence group</i>		
A6	Remove semantic entropy from $\hat{u}$	Does SE uniquely contribute to hallucination detection?
A7	(q, a) -only training (no reasoning chain c)	Does reasoning-chain supervision improve over answer-only supervision?

Table 7: Extended ablations (new experiments on cloud GPU).

ID	Config	Est. cost	Purpose
PUB-04+05	Four-variant FT comparison: (i) full FT + L2, (ii) full FT + EWC, (iii) LoRA ($r=8$, $\alpha=16$) + L2, (iv) LoRA alone	~ 8 h	Validates L2 as correct EWC approximation (PUB-05) and pre-empts “why not PEFT?” question (PUB-04). If LoRA + L2 matches full FT + L2 on EM with better MMLU retention, recommend as future deployment path.
PUB-06	Flan-T5-XL (3B) on FEVER only	~ 6 h	Tests whether CAEM mechanism holds at larger scale. Also partially redeems the HotpotQA capability-ceiling finding: if 3B clears the $p > (1 - \alpha)$ precondition on HotpotQA, confirm as a note.
PUB-03	Human evaluation on TruthfulQA (N=100)	No GPU	Two annotators evaluate Cycle 0 vs. Cycle 10 output pairs blind. Report Cohen’s κ . Validates ROUGE-L proxy used in automated evaluation.

6.3 What to Drop

Full-scale HotpotQA (780M) run — dropped. EXP-19 already confirmed $EM \approx 0.006$ across 4 cycles at $n=500$. Scaling to $n=5,000$ on a benchmark the model cannot answer adds no new information. Report EXP-19 mini-run data as the boundary condition finding in Chapter 5. This saves ~ 8 – 10 GPU hours.

7 Labels and Scientific Validity

This section records the final position on ground-truth dependency — the answer to the supervisor’s question.

Supervisor response (canonical): “CAEM uses approximately 500 labels per training benchmark for a one-time confidence calibration step. These labels calibrate *how much to trust the model’s confidence signal*, not what the correct answers are. After calibration, the self-improvement loop runs indefinitely on self-generated, self-verified data with no additional labels. A direct fine-tune on 500 samples would give a marginally better static model with no confidence signal, no routing, and no continued improvement. CAEM’s 500 labels buy a self-improving loop that runs across 10 cycles and transfers to held-out benchmarks the model was never trained on.”

Table 8: Ground-truth label dependency by component.

Component		Labels needed?	What it uses instead
NLI (Layer 1)	verification	No	Retrieved Wikipedia passage as premise; generated answer as hypothesis. Self-supervised.
Self-consistency (Layer 2)		No	$M=3$ model outputs compared against each other.
Semantic (Layer 3)	entropy	No	$K=10$ samples; NLI-based equivalence clustering.
Self-improvement tuning	fine-	No	Trains on model’s own verified (passage, reasoning, answer) triples.
$\hat{u}$ weight calibration		~ 500 samples	Logistic regression on (signal values, correct/incorrect) pairs. One-time bootstrap; no ongoing label dependency.
Cold-start seeding		Optional	Can use existing QA datasets or self-generate + verify seed episodes.
Evaluation		Yes (benchmark)	Ground-truth labels used for EM/F1 measurement only — not part of the training loop.

Table 9: Chapter writing status and dependencies.

Chapter	Content	Status	Blocking dependency
Ch 1	Introduction	DONE	None
Ch 2	Literature Review	DONE	None
Ch 3	Requirements, Impact, Project Management	DONE	None
Ch 4	Methodology (full)	DONE	Add formal purity theorem proof (PUB-02, $\sim 2h$, no data needed)
Ch 5	Experimental Results	NOT WRITTEN	Needs full-scale cloud experiment results
Ch 6	Conclusion	NOT WRITTEN	Needs Ch 5 complete

8 Writing Plan

8.1 Chapter 5 Structure

1. **§5.1 Experimental Setup** — hardware (A100 cloud), datasets (Table ?? + Table ??), baselines (Table ??), evaluation metrics (EM/F1, HallRed%, MMLU retention).
2. **§5.2 Main Results** — per-cycle accuracy table across all training and transfer benchmarks; primary narrative thread: “Does CAEM outperform baselines lacking a closed loop?”
3. **§5.3 Mechanism Evidence** — three evidence tables: (a) Tier 1% growth per cycle (memory accumulation); (b) $\bar{u}_{\text{stored}}$ rising + stored% falling per cycle (memory quality); (c) retroactive re-verification delta per cycle (backward purity improvement).
4. **§5.4 Transfer Evaluation** — TruthfulQA, StrategyQA, ARC-Challenge results post-Cycle 10. Headline: generalised hallucination reduction without task-specific adaptation.
5. **§5.5 Theoretical Validation** — purity theorem empirical check; convergence curve vs. theoretical prediction; HotpotQA boundary condition (mini-run data).
6. **§5.6 Ablation Study** — internal ablations (Table ??) + extended ablations (Table ??). Each ablation answers one specific committee question.
7. **§5.7 Efficiency and Stability** — per-tier latency, Tier 1 fraction growth (latency dividend), MMLU retention across cycles, forgetting abort activations (if any), total GPU hours.
8. **§5.8 Error Analysis** — HotpotQA capability ceiling (theory validation); TruthfulQA ROUGE-L instability; human eval agreement (PUB-03).

8.2 Chapter 6 Structure

1. **§6.1 Summary of Findings** — one paragraph per confirmed mechanism + one for the HotpotQA boundary condition.
2. **§6.2 Limitations** — model scale dependency (780M ceiling); calibration label requirement (~ 500 samples); inference scope (NLI requires retrieved context in open-domain mode); ROUGE-L unreliability for TruthfulQA; single-GPU reproducibility claim updated to A100 cloud.
3. **§6.3 Future Work** — (i) model capacity scaling (Flan-T5-XL/XXL); (ii) knowledge distillation from verified memory into smaller student models; (iii) reference-free NLI using retrieved passages instead of benchmark labels; (iv) LoRA fine-tuning path for deployment efficiency (from PUB-04+05 if it matches full FT).

9 Execution Sequence

Parallel track A — RTX 3060, start immediately (no cloud needed):

1. Apply FEVER leakage fix to `run_experiment.py` and `caem/benchmarks/fever.py`.
2. Re-run mini smoke test ($n=50$, 2 cycles) to confirm FEVER results survive the fix.
3. Run `scripts/run_ablation.py` → MMLU retention numbers (needed for Ch 5 §5.7).
4. Run `scripts/run_purity_validation.py` → Theory 1/2/3 validation tables (needed for Ch 5 §5.5).
5. Write formal purity theorem proof block in Chapter 4 (PUB-02, ~2h, zero compute).
6. Update `config.py`: `max_memory_size = 1_000_000`, `n_cycles = 10`, `faiss_nlist = 4096`.

Parallel track B — Cloud A100 80 GB setup:

1. Add TriviaQA and Natural Questions benchmark adapters (`caem/benchmarks/triviaqa.py`, `caem/benchmarks/natural_questions.py`).
2. Add StrategyQA and ARC-Challenge as transfer-eval-only adapters.
3. Download and build full DPR Wikipedia FAISS index (21M passages; build once, reuse across all cycles).
4. Run `scripts/seed_cold_start.py` for FEVER + TriviaQA + NQ (~600 total seed episodes).

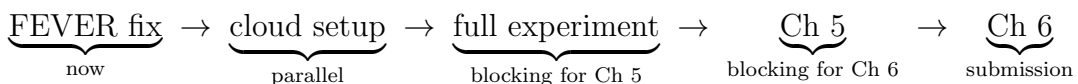
Main cloud experiment (after tracks A + B complete):

1. Full-scale run: FEVER + TriviaQA + NQ, 10 cycles, 5,000/benchmark/cycle, 1M memory, full DPR Wikipedia.
2. Post-cycle transfer eval: TruthfulQA, StrategyQA, ARC-Challenge (zero-shot, no training).
3. PUB-04+05: 4-variant fine-tuning ablation (full FT + L2 / + EWC; LoRA + L2 / alone).
4. PUB-06: Flan-T5-XL (3B) on FEVER, 3 cycles only.
5. PUB-03: Human eval on TruthfulQA 100-sample subset (no GPU).

Writing (after main experiment):

1. Chapter 5 (all sections, full data available).
2. Chapter 6 (findings + limitations + future work).
3. Appendices (extended ablation tables, XL model results, full per-question analysis).

9.1 Critical Path



The FEVER fix is the single earliest blocking item. Do not start the full-scale cloud

run until the fix is applied and smoke-tested.